

VATER RULES

VATER RULES — the numbered rulebook (compiled 2026-08-12 • Section Two 2026-08-15 • Section Three 2026-08-16)

Every standing rule Trey/Jared has given for how Vater videos are created and operated, in one place, numbered for quick reference (“rule 47”).

Numbering doctrine (same as the library #N system): rule numbers are PERMANENT — never renumbered, never reused. New rules get the next free number and are appended to their section. Retired rules keep their number, struck through with a date.

Authority chain: machine truth is `VATER-STANDING-SPEC.json` (read by `vater.py` and the finance-pipeline harnesses via `finance-pipeline/standing-spec.mjs`). This file is the human-readable layer. **To change a rule:** edit or annotate it here — the next Vater session syncs the change into `VATER-STANDING-SPEC.{md,json}` (and `VATER-SETTINGS.env` where applicable). Each rule cites its source so drill-down is one hop.

1. Intake, script & approval gates

1. **Script-first.** The source of truth is TREY’S SCRIPT. Reference videos are RETIRED — no yt-dlp, no whisper transcription, no LLM scripting pass, and no spend of any kind until Approve. (*VATER-STANDING-SPEC.md §2*)
2. **Script Review flow:** Trey pastes/uploads → reads/edits/saves → **Approve & Animate.** New rows park at `awaiting_script_approval` via `POST /api/vater/youtube/from-script`. (*vater-script-first-jeff-whitfield*)
3. **Nothing renders before approval.** `stopAfterScript` → `script_ready` → `awaiting_script_approval` → re-kick with `scriptOverride`. (*vater-video-qa-doctrine*)
4. **Script saves are versioned, append-only** (`YouTubeProject.scriptVersions`: generated/edited/approved, cap 20). (*vater-standing-spec*)
5. **Draft-gated output.** No publishing, no YouTube upload, no customer handoff without Trey/Jared pulling the trigger. (*vater-sandbox-doctrine, vater-direct-lane*)
6. **Fact-check the script stage.** Kimi hallucinated “avg card debt \$1,000” (real ≈ \$6.5k) — numbers in scripts get verified. (*vater-finance-pipeline*)
7. **Runtime comes from the script, never from stretching scenes.** Want a longer video → write more words. Target words ≈ minutes × 185 (Monroe long-form). (*vater-video-qa-doctrine, SPEC §6*)
8. **A missing locked style is a hard 409, never a silent fallback.** (*vater-script-first-jeff-whitfield*)
9. **DO-NOT-SHIP: “wealth-in-30s”** — video + short stay in the library; no publish/upload/staging until Jared reverses. (*vater-finance-pipeline*)
10. **#21 “Buy vs Rent”** (`cmsoxr9od000114ng7geu9hqo`) — rejected by Trey 8/12; repair SHIPPED 8/13 (93 slides remade against benchmark #19, recomposed to the same blob key, \$13.27 all-in). Captions/wpm fixes still open; publishing still waits on Trey/Jared per rule 5. (*vater-library-numbering*)

11. **Course lane gates:** Approve & Render in Course Studio is the structural gate; hard STOP after the Lesson-1 pilot for quality+cost review before any batch. (*vater-financial-course*)

2. Style & look

12. **ONE locked style: “Jeff Whitfield 3-D style”** (formerly VATER-3D / Animate2-Modal). The site’s style picker is GONE; Script Review binds to this row via `lib/vater/locked-style.ts` (DB `cmofyvao30000l204op279f52`). (*SPEC §1*)
13. **Stills** = preset `pixar` + custom art style **“Finance Pixar 3D”** (`finance-pixar-test`). (*SPEC §1*)
14. **NEVER 2D**. Persistent default, not a per-job argument. (*SPEC §1*)
15. **NEVER photoreal / live-action / realistic humans** — every scene stays stylized 3-D cartoon. (*SPEC json neverPhotoreal*, Trey 8/10)
16. **Motion engine:** Wan2.2 **Animate-2 on Modal** (`vater-wan22.animate2_clip`, L40S, distill Q8, 6-step, 16fps native → 32fps compose, 81 frames/chunk max); fallback `modal-wan22` for scenes with no usable driver. (*SPEC §1*)
17. **Every still scene gets camera motion — never a frozen frame**. Four directions (zoom in / zoom out / pan L → R / pan R → L) cycling so consecutive scenes never repeat; smooth, no jitter; 4× supersample before zoompan and don’t lower it (ZOOM_MAX 1.12, PAN_ZOOM 1.08). (*SPEC json stillMotion*, Trey-approved 8/10)
18. **Default cost posture = STILLS + PANNING; animation is opt-in per video, never the default** (still ≈ \$0.05 vs animated clip ≈ \$0.50). (*SPEC json costPosture*, Trey 8/10)
19. **Hard cuts only — no fading between scenes** (`VATER_SCENE_XFADE_S=0`, tail trim 0.20s); keep the single 1.5s end fade. (*vater-video-qa-doctrine*, *VATER-SETTINGS.env*)
20. **Quality is locked** at 848x480 @ 32fps, NVENC p4 cq23 (~1.1 Mb/s), libx264 crf20 fallback, AAC 192k 48kHz, 4× still supersample, no frame interpolation. Changes need explicit approval. (*SPEC §8*)
21. **Captions are LOCKED ON exactly as they are** — style, size, position, timing untouched without Trey’s explicit approval. (*SPEC json captions*, Trey 8/10)
22. **Animated clips need a camera push** (`VATER_ANIM_PUSH=0.12`) or they read as slides; judge “does it look animated” by within-scene frame diff 2s apart, never across a cut. (*vater-video-qa-doctrine*)

3. Cast & characters

23. **Host = Jeff Whitfield, in every video**, the primary character explaining the finance content. Assets `characters/jeff-whitfield/`; seed 7 (`VATER_BASE_SEED`); no LoRA — identity = descriptor + pinned seed + `cast_match QA`. (*SPEC §3*)
24. **Closed cast**. The only humans allowed anywhere: Jeff Whitfield, Linda Whitfield, David Reyes, Cooper Shaw, Julie Tran, Lynette Brooks — and only in scenes the script names. No tellers, colleagues, passers-by, crowds, background figures, or unnamed hands. A scene naming no roster character has NO people at all. (*SPEC json castRules.closedCast*)
25. **Script-named roster characters MUST appear; unnamed ones must not**. `lockRoster` turns auto-casting off. (*SPEC json characterLock*)
26. **NEVER two of the same character in one frame** (no Jeff looking at Jeff) — enforced in planner prompt, deterministic lint, and the `no_clones QA` check. Two *different* characters sharing a frame is fine. (*SPEC §3c*)

27. **ONLY A HUMAN HAS A FACE.** Piggy banks, cards, calculators, jars, coins are inanimate — never eyes/mouths/limbs, never cute or anthropomorphized. (*SPEC json facesRule*)
28. **No faced objects for counts** — never piggy banks / moneyboxes / mascots / figurines (proven unfixable); use bank cards, passbooks, labelled glass jars, envelopes, statements, cash stacks, on-screen account cards. Enforced by `_FACED_OBJECT_RE`. (*SPEC json noFacedObjects*)
29. **Identity lock.** Every character keeps identical face, hair (color/length/style), facial hair, eye color, skin tone, freckles, earrings, and build in every scene of every video. ONLY clothing and setting vary. (*SPEC json characterLock, Trey 8/10*)
30. **Per-character invariants are explicit:** David = tightly-curved black hair, always bearded; Cooper = copper-tinger wavy hair, always bearded, green eyes, freckles; Julie = always high ponytail, silver studs, no freckles; Lynette = shoulder-length cornrow braids, gold hoops. (*SPEC json supportingCharacters*)
31. **Descriptors state locked features as POSITIVE concrete description, never negations** — “never down” produced hair down. (*SPEC json descriptorWritingRule*)
32. **Wardrobe = one outfit per ACT** (up to 3 acts/video), seeded by (character, jobId) so resumes re-pick the same outfit; never per-scene randomness. Clothing was removed from identity descriptors so it can vary. (*SPEC §3a*)
33. **Jeff is a financial businessman — suits or professional attire, ALWAYS.** Palette characters/jeff-whitfield/wardrobe.json. (*SPEC §3a*)
34. **The host stays index 0 in any roster** (vater.py treats characters[0] as host); Prisma character includes have no ORDER BY — sort oldest-first. (*vater-script-first-jeff-whitfield*)
35. **NEVER clone a real person’s voice or use real-person photo refs.** (*vater-video-qa-doctrine*)

4. Scenes, backgrounds & on-screen text

36. **Backgrounds MUST VARY: ≥3 distinct settings per video**, no single setting over $\sim\frac{1}{3}$ of scenes, interiors + exteriors mixed; the narration picks the place. Palette SETTINGS-PALETTE.json. (*SPEC §4, Trey-approved 8/10*)
37. **On-screen text MINIMAL — numbers only**, matching what the script is saying at that moment; never paragraphs, headlines, signs, labels, or characters holding written text. (*SPEC §7*)
38. **\$-header / drawtext cards stay OFF** (VATER_FINANCE_HEADERS=off, Trey 8/8). Manual editor overlays still render. (*vater-video-qa-doctrine*)
39. **KEEP and encourage charts, graphs, ledgers, calculators, and financial tools Jeff is using** — clean and near-unlabeled. More financial imagery overall. (*SPEC §7*)
40. **More financial props per scene, but props carry little to NO readable text** (22-item palette in SPEC json propsPalette). (*Trey 8/10*)
41. **Props must be motivated by that beat’s narration. NO symbolic/surreal props ever** (brain models, word-globes, badges, trophies) — `_strip_surreal_props`. (*vater-video-qa-doctrine*)
42. **Visuals land in the SAME scene as their narration — never one beat late.** (*vater-video-qa-doctrine*)
43. **Expressions match the narration’s emotional tone per scene** (planner emits emotion from a closed vocabulary). (*vater-video-qa-doctrine*)
44. **Visual vocabulary rule:** when the narration states a COUNT, NUMBER, LIST, COMPARISON, or FLOW — SHOW it, usually without the host (“eight bank accounts” → eight countable cards; “4.9% vs 0.01%” → both figures large, side by side). (*SPEC json visualVocabulary*)

45. **At least 1 scene in 3 is a people-free financial visual** — and these visuals are the ONE place on-screen text (figures, short labels, list lines) is encouraged. (*SPEC json minimumShare/textException*)
46. **BIG props:** a screen or phone filling about half the frame, shot close, clearly readable, showing the figure the narration just said. Digits render reliably; long words render as gibberish. (*SPEC json bigProps/textOnProps*)
47. **One labelled container (jar) per named fund** the narration mentions. (*SPEC json labelledContainers*)
48. **Banned filler poses:** pointing off into the distance, two fingers / peace sign / thumbs up at camera, staring at a wall calendar, gazing at an unspecified paper. (*SPEC json bannedFillerPoses*) — 8/16: any finger-COUNT gesture (“three fingers raised”, “two fingers extended”) renders as a V-sign; stage counts on calendars/tablets, never on hands. (*vater-59-and-a-half-render*)
49. **Host in ~half of all scenes**, doing something that matches the narration; the other half people-free financial visuals. (*SPEC json hostPresence*)
50. **Never a scene that contradicts the narration** (broken car window with no car in the script; Christmas ornaments during sinking funds). (*SPEC json benchmarkVideo.fix*)
51. **Strip anatomy clauses from the style suffix on people-free scenes** (`_scene_style_suffix`) — that suffix is what put faces on objects. (*SPEC json rootCause2026_08_10*)

5. Voice & audio

52. **Voice = Monroe** — local IndexTTS-2 clone (`vater_voices/Monroe.wav`) served via Modal app `vater-tts` (`voiceBackend: "indextts-modal"`). (*SPEC §6, vater-modal-only*)
53. **The TTS Modal path NEVER falls back to f5-tts** — that fallback silently re-enters the local GPU. (*vater-modal-only*)
54. **Approved tempo:** `VATER_TTS_INTERVAL_MS=340`, `VATER_TTS_CHUNK_GAP_MS=320`, `VATER_TTS_TEMPO=0.94` (~155 wpm delivered). (*SPEC json voice.pacing*) — **superseded per voice by a locked Voice Tuner file (rule 138)**.
55. **Measure wpm per voice** — Monroe reads ~185 wpm long-form / ~164 short; short samples mislead. (*SPEC §6, vater-video-qa-doctrine*)
56. **Tone = calm, authoritative, human, unhurried — explaining, not selling** (Nick Invests feel). (*vater-video-qa-doctrine*)
57. **Punctuation IS the tempo.** Scripts are written punctuation-for-the-ear; an em dash produces a long beat in IndexTTS-2. The ref clip’s own pace is the real lever; sampler/emotion params barely move it (± 3 wpm). (*vater-video-qa-doctrine, vater-finance-pipeline*)
58. **Dollar amounts spelled out in words in the script AND voiceover** (“eight hundred thirty-nine thousand dollars”, never “\$839k”); `speakable.py` handles legacy digit scripts. ⚠ See conflict C7 (captions want numerals). (*vater-video-qa-doctrine*)
59. **Acronyms spelled out in the spoken read only:** IRA → “I-R-A”, FDIC → “F-D-I-C” (`_spell_finance_acronyms`); captions keep “Roth IRA”. (*SPEC json pronunciation*) 8/16: map extended with IRAs, RMD/RMDs, IRS, CDs.
60. **Lip sync OFF.** Mouths neutral or closed; S2V stays per-scene ask-only; don’t spend compute on it. (*SPEC §5*)
61. **Final mux pins -ar 48000 -ac 2.** (*vater-blob-delivery*)

6. QA & pre-ship

62. **Still-QA runs on every still AFTER stills, BEFORE Modal animation spend** (`vater_scene_qa.py`, Gemini vision): `no_clones`, `person_count_ok`, `cast_match`, `props_motivated`, `no_readable_text`, `anatomy_ok`, `no_ghost_figures`, `setting_props_consistent`, `prop_type_match`, `no_self_regard`. `VATER_SCENE_QA=on`. (*vater-video-qa-doctrine*)
63. **Clip-QA** (`vater_clip_qa.py`) judges 3 frames per animated clip (30/60/90%) before compose; fail → one re-animate → revert to Ken Burns still. (*vater-video-qa-doctrine*)
64. **Final pre-ship step is a human/Claude EYEBALL of extracted frames (60% + 90%) of every animated clip** — Gemini flash misses subtle extra hands at 480p and false-flags cartoon finger counts. (*vater-video-qa-doctrine*)
65. **Always extract frames from the FINAL at the fixed scenes' timestamps to confirm a revision landed** — never trust "REVISED + RECOMPOSED". (*vater-video-qa-doctrine*)
66. **Gate QA on defects you can point at**; `expression_ok` is ADVISORY only. Re-roll budget: 2 per scene, 20 global. (*vater-video-qa-doctrine*, `VATER-SETTINGS.env`)
67. **Always eyeball crowd / "most people" beats** — they render as 2–5 identical clones and still-QA passes them; fix via single-host or prop staging. (*vater-standing-spec*)
68. **Never stage prop-only beats without the host** — FireRed grows faces on prop-only scenes. (*vater-standing-spec*)
69. **Production-ready standing order**: every video is built to be publishable without Jared re-reviewing — self-check anatomy, numbers, timing, tempo before delivering. (*vater-video-qa-doctrine*)
70. **Trey's/Jared's scene numbers in reviews are 1-BASED** (his "scene 3" = `scenes/002.png`). (*vater-video-qa-doctrine*)
71. **Motion check**: average motion score < 15 (mean pixel diff) → STOP, the prompts didn't unlock motion. Target 15–25 for narrative. (*vater-animation-recipe*)
72. **Never kick parallel /animate-scene calls for the same scene index** (version-numbering race, unfixed). (*vater-animation-recipe*)

7. Backend & infra

73. **HARD RULE: Vater is MODAL-ONLY. Never run any part of a Vater render on the DGX GPU.** ("never use Vater with GPU only send it to Modal" — Jared 8/10) (*vater-modal-only*)
74. **NEVER `systemctl --user start/restart comfyui-vater`** (or any ComfyUI) to unblock a Vater job — route it to Modal instead. (*vater-modal-only*)
75. **All four lanes are Modal**: stills `firered-modal`, animation `modal-wan22`, planner/script LLM off-box via `LiteLLM:4000` → `fallback/kimi-k2-turbo`, voice `indextts-modal`. (*vater-modal-only*)
76. **wan22-local on the GB10 is explicitly rejected** (15–20 min/clip — "a 30 sec clip that takes 3 hours is not practical"). (*vater-finance-pipeline*, Jared 8/5)
77. **Runway-clear / stopping other prod services gates on real local-GPU use (`uses_local_gpu`)**, never on `videoBackend == "sdxl"`; keep the flag and `schedule_restore` in sync. (*vater-modal-only*)
78. **One general env-driven harness**: `~/vater-studio/finance-pipeline/vater-render.mjs` — no more fork-per-video scripts. (*vater-modal-only*) 78a. **TTS chunk concat pads in WHOLE FRAMES** (`_run_indextts_modal`, tuner preview path): Modal returns 22,050 Hz s16 mono; a byte-count pad that lands ODD (450 ms gap) shifts every second chunk one byte → broadband noise. Fixed 8/16. (*vater-59-and-a-half-render*)

- 79.  **systemctl --user restart content-autopilot-api** RESUMES every job stored as running — before restarting, check `vater_jobs.json` and `POST /vater/cancel-job` on each; gate the restart on the check output, never chain check && restart in one command. (*vater-script-first, vater-financial-course*)
- 80. **Never default to `sdxl-local` / `sdxl-ipadapter` / FLUX for scene gen.** (*feedback_vater_no_sdxl*)
- 81. **Modal wan batches chunk at `VATER_ANIM_CHUNK_SCENES=12`; a DGX cancel does NOT kill Modal tasks** — the function timeout is the only real kill. (*vater-finance-pipeline*)
- 82. **The poll route's done handler must MERGE `scenesJson` per `idx`, never overwrite** (overwrite wipes `mediaType/videoUrl/videoVersion/animQuality`). (*feedback_vater_poll_merge_scenes*)
- 83. **Vater Watcher runs on a scoped allowlist** (5 MCP tools + `Read/Grep/Glob`, no `Bash/Edit/Write`); real changes only via `propose_action` → user clicks Apply. (*vater-watcher*)
- 84. **Long-form renders use a 15-min AbortSignal on fetch** (cold Modal regen exceeds undici's 5-min default). (*vater-standing-spec*)
- 85. **Never hardcode style/character/settings in a harness** — import `finance-pipeline/standing-spec.mjs` or read the `SPEC.json`; any spec change lands in `VATER-STANDING-SPEC.{md,json}` FIRST. (*vater-standing-spec*)
- 86. **Never hardcode secrets** — `set -a; . ~/vater-studio/VATER-SETTINGS.env`. `CONTENT_API_KEY` truth = the content-autopilot-api systemd unit; never a code default; never in a client bundle. (*content-api-key-rotation*)

8. Library & delivery

- 87. **Permanent #N – title prefixes** on every library video (per owner, oldest-first); numbers NEVER change or get reused — Trey references videos by number. Legacy ownerless rows use `#LN` →. Enforced by `lib/vater/video-number.ts` + lazy sweep on every library fetch. (*SPEC §9*)
- 88. **Never create or rename a Vater video without keeping the #N – prefix intact.** (*vater-library-numbering*)
- 89. **The approved format benchmark is #19 — The Three-Account System (v3)**, project `cmsolefbqb0001l446nhl1psbl` (Trey corrected the pointer 8/12; his 8/11 approval was for v3). Never cite the benchmark by title alone — four renders share it (#15 v1, #18 v2, #19 v3 = APPROVED, #20 v4). (*SPEC §9 + json benchmarkVideo*)
- 90. **Every render/example goes to the studio Library + green NEW banner push** (`/api/vater/latest`) — never NAS-only. (*vater-sandbox-doctrine*)
- 91. **Finals serve from Vercel Blob CDN** (`vater-finals/<projectId>.mp4, ?v=<ts>`); playback links straight to `*.blob.vercel-storage.com` via `finalVideoPlaybackUrl()`, never the serverless proxy. Download stays on the proxy (`?download=1`). (*vater-blob-delivery*)
- 92. **The poll route must NEVER downgrade an `https finalVideoUrl` back to a proxy path**; any new run script writes the blob URL. (*vater-blob-delivery*)
- 93. **Compose always writes `-movflags +faststart`** (moov at front). (*vater-blob-delivery*)
- 94. **Audio-but-no-video / stutter for external users is a DELIVERY problem** — never re-litigate codecs for that symptom class. (*vater-blob-delivery*)

9. Cost, billing & payments

- 95. **The per-video all-in total is part of the deliverable** — never report a video done without its cost; never make Jared ask. (*vater-cost-truth-automated*)
- 96. **Costs are auto-reconciled from the job record** (`scripts/vater-cost-reconcile.mjs`, cron every 20 min, log `~/vater-studio/cost-reconcile.log`) — never hand-total a video. (*vater-cost-truth-automated*)
- 97. **DB byJob is the durable ledger; file amounts MERGE over it, never rebuild.** A total going DOWN with fewer jobs = rotated history, not a correction. (*vater-cost-truth-automated*)
- 98. **Never hand-edit costJson without adding the jobIds to byJob.** (*vater-cost-truth-automated*)
- 99. **Apply VATER_MODAL_OVERHEAD_MULT=1.6 at cost finalize;** if reconciliation still gaps, tune the mult — don't stack rows. (*VATER-SETTINGS.env*, *vater-finance-pipeline*)
- 100. **Invoicing = exactly two lines:** Compute (at cost, passed through unchanged) + Render Operations (finished minutes × rate). (*vater-billing-ops-fee*)
- 101. **VATER_OPS_RATE_PER_MIN = 0.35;** every code default must equal 0.35 — NEVER reintroduce a 1.00 fallback. (*vater-billing-ops-fee*)
- 102. **ONE NUMBER on the tab** = Compute (all cash, at cost) + Render Operations; click-through shows the FULL provider breakdown as plain one-liners — never strip the provider list, no parentheses, no rate math. (*vater-billing-ops-fee*)
- 103. **A running total must NEVER shrink** — the pill shows \$X spent · \$Y billed. (*vater-billing-ops-fee*)
- 104. **Token counts are DEAD on all cost surfaces** — dollars and stage breakdown only. (*vater-billing-ops-fee*)
- 105. **Cost dollars = REAL cash only** (Modal/Gemini/fal); Claude tokens never convert to dollars. (*vater-finance-pipeline*)
- 106. **Local-first cost discipline:** DGX \$0 → Modal cents → FLAG before any new paid API (incl. fal.ai). (*vater-sandbox-doctrine*)
- 107. **Billing/payment totals math lives ONLY in lib/vater/billing/summary.ts** (`getVaterBillingSummary`); never fork it. (*vater-payments-due-reset*)
- 108. **recordVaterPayment()** is the ONE write path for payments (a bare `prisma.vaterPayment.create` loses the snapshot). (*vater-payments-due-reset*)
- 109. **All-time total NEVER resets; dueUsd = totalUsd – payments.** Reset when the weekly Zelle lands, via `/hq "Trey render bill" or node ~/vater-studio/vater-payment.mjs <amt> --note "..."`. (*vater-payments-due-reset*)
- 110. **Never record a payment before the Zelle actually lands** (hard-money rule). (*vater-payments-due-reset*)
- 111. **The ops fee is REPORTING only** — live Stripe charging is untouched; wiring it needs Jared's explicit word. (*vater-billing-ops-fee*)
- 112. **The cost pill stays all-in across BOTH lanes** (YouTube + course) — never filter the `/api/vater/latest` rollups. (*vater-financial-course*)

10. Scope & sandbox

- 113. **"Vater" scopes a conversation EXCLUSIVELY to the /animate studio + its plumbing** — nothing else gets changed, added, edited, or even verified. (*vater-sandbox-doctrine*)
- 114. **In-scope surface:** `tolley.io/animate`, `app/api/vater/*`, `components/animate/**`, `components/vater/**`, `lib/vater/**`, DGX content-autopilot-api :8096 vater routes, `comfyui-vater :8189` (dormant per rule 73), `vater-priority`, `claude-vater-watcher`, Modal apps `vater-wan22/vater-firered/vater-tts`, `~/vater-studio/`. (*vater-sandbox-doctrine*)

- l15. **Broken non-Vater things get FLAGGED to Jared, never fixed inside a Vater conversation.** (*vater-sandbox-doctrine*)
- l16. **Trey's YouTube OAuth tokens go to vater-scoped storage** (SocialAccount via /api/vater/social-accounts/oauth/youtube/*) — NEVER PlatformConnection (Jared's channels). (*vater-sandbox-doctrine*)
- l17. **Studio-only access = VATER_STUDIO_ALLOWLIST_EMAILS only** — NEVER VATER_ADMIN_ALLOWLIST_EMAILS (that flag exposes Plaid/net-worth data; live breach 2026-08-05). (*vater-studio-access-tier*)
- l18. **Verify access changes against prod env** (npx vercel env pull), never local .env. (*vater-studio-access-tier*)
- l19. **The Direct lane runs as Unix user vater** (uid 1001, no sudo/docker, ACL-sealed to ~/vater-studio); all renders via the :8096 API; finance scripts stay owner-side. (*vater-direct-lane*)

11. Director rules — Section Two (Trey 2026-08-15, applies to ALL videos)

Second training batch from Trey. Where a rule sharpens an earlier one, the earlier number is cited; the new number is the one to reference going forward.

- l20. **Anatomy: every character has exactly TWO arms and TWO hands** — clearly separate from each other and from any object; no extra, fused, or hidden-then-duplicated limbs. Enforced by the anatomy_ok QA check + re-roll. (*Trey 8/15*)
- l21. **A car is NEVER inside a room.** If the script mentions a car it lives in a driveway, garage, street, lot, or repair bay — never in an office, kitchen, or living room. (*Trey 8/15; sharpens rule 50*)
- l22. **“Truck” = a PICKUP truck** — unless the script literally says “moving truck” (then a box/moving truck). Never default to a moving van. (*Trey 8/15*)
- l23. **No bland prop scenes.** A financial scene is never just random pots, vases, or décor. Every people-free scene is a clear FINANCIAL visual — cards, statements, cash, ledgers, charts, calculators, screens, jars — as the obvious subject. (*Trey 8/15; sharpens rules 41, 45*)
- l24. **ONE cartoon feel for the whole video.** People-free/prop-only scenes are rendered in the SAME stylized 3-D cartoon look, lighting, palette, and polish as the scenes Jeff is in. A prop scene must look like it was cut from the same movie — never flat, photoreal, or “stock”. (*Trey 8/15; sharpens rules 14–15*)
- l25. **Scene-to-scene consistency** — even when no character is on screen, the style, colour grade, and world stay continuous with the character scenes. (*Trey 8/15*)
- l26. **Never duplicate a main character in a scene** — one Jeff, one Linda, etc. Reaffirms rule 26 (no clones / no Jeff looking at Jeff). (*Trey 8/15*)
- l27. **Scene lengths VARY to match the tempo of the script** — never a fixed clock. (*Trey 8/15; = SPEC §10 sceneTiming*)
- l28. **Scenes change when a NEW SENTENCE starts** and follow the flow of the script. (*Trey 8/15; = SPEC §10 sceneTiming*) — **LIVE in the pipeline 8/16:** vater.py _schedule_scenes snaps every scene boundary to the nearest sentence start ($\pm 2.0s$, env VATER_SENTENCE_SNAP_S), else phrase start ($\pm 0.8s$), else word onset; floor 2.0s (VATER_MIN_SCENE_S). Scene count unchanged; the planner now sees one sentence per beat. (*vater-59-and-a-half-render*)
- l29. **What is on screen matches what the script is saying RIGHT THEN** — never lagging, never ahead, never unrelated. (*Trey 8/15; sharpens rules 42, 50*)

- l30. **Financial props are LARGE ENOUGH TO READ**, and any number on them ALWAYS matches the number the script is saying in that beat — digit for digit. *(Trey 8/15; sharpens rule 46)*
- l31. **Text on financial props = large, correctly-spelled ENGLISH words only**. Never gibberish, never pseudo-letters, never a foreign script. If the model cannot spell it, design the prop without the word. *(Trey 8/15; sharpens SPEC §10 textRule)*
- l32. **Characters never hold random signs with words on them**. *(Trey 8/15; reaffirms rule 37)*
- l33. **No peace sign / V-sign, ever** — no two-fingers gesture with hands or fingers. *(Trey 8/15; reaffirms rule 48)*
- l34. **All scenes share the same cartoon feel** — one locked style across characters, props, and backgrounds. *(Trey 8/15; reaffirms rule 12)*
- l35. **Financial props face the AUDIENCE**. A screen, statement, chart, or card is turned so the viewer can see its face — never edge-on, never showing its back. *(Trey 8/15)*
- l36. **Computers and phones have a screen ONLY on the front, visible side**. No screens on the back of a laptop lid or the rear of a phone; a device seen from behind shows a plain closed case. *(Trey 8/15)*
- l37. **Prompt craft for the above stays POSITIVE-ONLY** — render prompts describe the correct thing (“pickup truck in the driveway”, “laptop open, screen facing camera showing \$1,180”), never the banned thing. *(Trey 8/15; = rule 31 + SPEC §10 promptCraft)*
- l38. **Voice Tuner is the truth for how a voice sounds (8/15)**. /animate → Voices → 🗣️ Voice Tuner. Per-voice `vater_voices/<Name>.tuning.json` (generation knobs + ffmpeg EQ/post chain) is written by “Lock in” and honored by every IndexTTS-2/Modal render of that voice; unlocked voices keep rule 54’s env pacing and no EQ. Previews and renders share ONE code path (`vater_voice_tuning.py`) — what Jared hears in the ~30s sample is what ships. Never hand-edit the env tempo/gap for a locked voice; edit it in the tuner. *(vater-voice-tuner)*

12. Director rules — Section Three (Jared 2026-08-16, feedback on #26 “The Yes Count”; applies to ALL videos)

Jared’s review notes on video #26. Where a rule sharpens an earlier one, the earlier number is cited.

- l39. **Two characters in a vehicle stay in the vehicle TOGETHER — in EVERY vehicle scene**. If the script puts Jeff and Linda (or any two named characters) in a car or truck together, every scene set in that vehicle shows BOTH of them, in their seats, until the script moves one of them out. Never a solo shot of the driver in a scene the script wrote as a shared ride. *(Jared 8/16; sharpens rules 26, 129)*
- l40. **Vehicle scenes are shot from the FRONT, through the windshield, travelling WITH traffic**. Camera is on the hood looking back through the front glass at the occupants; the vehicle is in its own lane, pointed down the road; the road, lane lines, other cars, and scenery visible through the windows are consistent with forward travel. Never a vehicle stopped or turned sideways across the road, never a side-on or three-quarter parked shot standing in for a driving beat. *(Jared 8/16; sharpens rules 121, 122)*
- l41. **Driver and steering wheel on the LEFT side of the car (US) — which through the windshield is the RIGHT side of the screen**. Driver’s seat + wheel = screen-right, passenger seat = screen-left. Prompts spell this out every time (“Jeff in the driver’s seat on the right side of the frame, steering wheel in front of him; Linda in the passenger seat on the left side of the frame”). *(Jared 8/16)*
- l42. **The cartoon style NEVER changes mid-video**. Every scene — host, secondary, prop-only, vehicle, exterior — is the same locked 3-D cartoon render, same lighting family, same palette, same polish. A style drift on even one scene is a re-roll, not a keep. *(Jared 8/16; reaffirms rules 12, 124, 125, 134)*

- l43. **ALL text shown ANYWHERE in ANY video is ENGLISH.** This extends rule 131 from financial props to everything on screen: signs, storefronts, screens, packaging, book spines, whiteboards, posters, street signs, dashboards, backgrounds. Non-English or pseudo-script anywhere in frame = re-roll; if the model cannot spell it, design the object without the lettering. (*Jared 8/16; sharpens rules 37, 131*)
- l44. **Finance-prop numbers on screen matched the script in #26 — keep doing that** (rules 46, 130 confirmed working; the number on the prop is the number being spoken, digit for digit). (*Jared 8/16, positive feedback — retained as a keep*)
- l45. **A video may run on an ElevenLabs voice by voice_id.** Harness: VOICE_BACKEND=elevenlabs
FINANCE_VOICE=<voice_id> VOICE_LABEL=<name>; narration is cloud (no local GPU, rule 73 intact); long scripts are chunked on paragraph/sentence boundaries under the 10k-char per-request cap; the spend is booked to the job as real cash (elevenlabs bucket, ELEVENLABS_USD_PER_1K_CHARS, ledger source elevenlabs) — never \$0. First use: #27 (voice UgBBYS2s0qTuMpoF3BR0 “Mark – Natural Conversations”, Jared 8/16). (*vater.py _run_elevenlabs_tts*)
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Appendix: Known conflicts & stale pointers (need a ruling or cleanup)

- **C1 — Benchmark #19 v3 vs #20 v4.** SPEC md+json (newest write, 8/12 18:39) say **#19 v3** (cms0lefqb0001l446nhl1psbl) and document Trey’s 8/12 self-correction (his 8/11 approval was v3). Two memory files and the JSON’s top-level authority string still said #20 v4 — memory now corrected; the JSON authority line is stale and awaits a spec-sync pass.
- **C2 — Standing spec v1 vs v2.** v1 (host “Ray Whitfield”, reference videos, style picker) is fully superseded by v2 (Jeff Whitfield, script-first, picker gone, wardrobe/backgrounds vary).
- **C3 — Reference-video residue.** SPEC json still carries referencePolicy: “subject-only”; the reused-reference 409 gate and script-from-reference route exist in code but are unreachable from the UI. Vestigial under v2.
- **C4 — VATER-SETTINGS.env still has FINANCE_CHARACTER_NAME=Ray Whitfield.** Possibly load-bearing in prompts — needs a verify pass + Trey’s ok before changing (output-affecting). Driver dir already repointed to jeff-whitfield.
- **C5 — Linda’s status.** SPEC md §3b says “pending critique”; json + REVISION-LOG say LOCKED (Trey 8/10, “Linda looks good”). The md paragraph is stale; David/Cooper/Julie/Lynette exist only in the json, not the md.
- **C6 — On-screen numbers have no mechanism.** Cards are OFF (rule 38) but the approved format wants the narration’s numbers on big props — today that depends entirely on the image model. Re-enabling Remotion counter_roll/stat_card_pop needs Trey’s yes.
- **C7 — Captions numerals vs words.** REVISION-LOG Round 1 locked captions to numerals (\$6,500) with speakable() for the TTS read only; QA doctrine rule 5 wants dollar amounts in WORDS in the script. Directly opposed — documented root cause of #21’s mangled captions (“ninety ,000”). Needs one ruling.
- **C8 — Caption style karaoke vs phrase blocks.** April doctrine said phrase blocks, “NEVER karaoke”; current pipeline ships karaoke captions, now LOCKED ON by Trey (rule 21). Current lock wins; also open whether “not a lot of text” covers captions — ASK BEFORE CHANGING.
- **C9 — Background variety vs story continuity.** The planner’s STORY-CONTINUITY rule beats the ≥3-settings rule on continuous scripts; continuity currently wins. Variety rule likely needs scoping per-act.

- **C10 — Kling/wan hybrid dormant.** The 8/8 Kling-first animation doctrine is dormant (not deleted) under the 8/10 stills-default cost posture; kling-pro is dead in code.
- **C11 — Transitions.** Round 2's 0.3s crossfade was reversed 8/8 → hard cuts only (rule 19). Hard cuts current.
- **C12 — Scene-gen backend.** April's firered-local default is superseded by firered-modal under Modal-only (rule 73).
- **C13 — Voice lineage.** Sterling → Wallace → **Monroe** (8/8). Any doc naming Sterling/Wallace as the finance voice is superseded.
- **C14 — wpm figures disagree** (185 long-form vs ~155 delivered at TEMPO 0.94 vs 164 short) — runtime estimation is the live weak spot; #21 was planned at a stale 155 while Monroe ran ~180.
- **C15 — vater.py still prefixes prompts with “A 2D cartoon male character.”** Deliberately untouched — it produced the look Trey approved; changing it counts as a quality change needing approval. The prefix drops when a named secondary shares the frame.